```
!pip install opencv-python numpy matplotlib -q

# Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt

print("Libraries imported successfully!")
print("OpenCV Version:", cv2.__version__)
print("NumPy Version:", np.__version__)
```

```
Libraries imported successfully!
OpenCV Version: 4.14.0
NumPy Version: 2.0.2
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files Screensho…234029.png

Screenshot 2026-08-19 234029.png(image/png) - 1172275 bytes, last modified: 8/19/2026 - 100% done

```
Saving Screenshot 2026-08-19 234029.png to Screenshot 2026-08-19 234029.png
```

```
# Get the uploaded filename
filename = next(iter(uploaded))

# Read image using OpenCV
image_bgr = cv2.imread(filename)

# Check whether image was loaded successfully
if image_bgr is None:
    print("Error: Image could not be loaded.")
else:
    print("Image loaded successfully!")
```

```
Image loaded successfully!
```

```
# OpenCV uses BGR format
# Convert BGR to RGB for correct display in Matplotlib
image_rgb = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8, 6))
plt.imshow(image_rgb)
plt.title("Original Color Image")
plt.axis("off")
plt.show()
```




```
# Image dimensions
height, width, channels = image_bgr.shape

print("Image Properties")
print("-----------------------------")
print("Width          :", width, "pixels")
print("Height         :", height, "pixels")
print("Number of Channels:", channels)
print("Image Shape    :", image_bgr.shape)
print("Data Type      :", image_bgr.dtype)
print("Total Pixels   :", width * height)
print("Total Values   :", image_bgr.size)
```

```
Image Properties
-----------------------------
Width          : 1131 pixels
Height         : 1032 pixels
Number of Channels: 3
Image Shape    : (1032, 1131, 3)
Data Type      : uint8
Total Pixels   : 1167192
Total Values   : 3501576
```

```
print("Image Resolution:", width, "x", height)
```

```
Image Resolution: 1131 x 1032
```

```
print("Minimum Pixel Value:", image_bgr.min())
print("Maximum Pixel Value:", image_bgr.max())
```

```
Minimum Pixel Value: 0
Maximum Pixel Value: 255
```

```
# Save image as JPEG
cv2.imwrite("output_image.jpg", image_bgr)

# Save image as PNG
cv2.imwrite("output_image.png", image_bgr)

print("Image saved successfully!")
```

```
Image saved successfully!
```

```
import os

jpeg_size = os.path.getsize("output_image.jpg")
png_size = os.path.getsize("output_image.png")

print("JPEG file size:", jpeg_size, "bytes")
print("PNG file size :", png_size, "bytes")
```

```
JPEG file size: 296237 bytes
PNG file size : 1155544 bytes
```

```
jpeg_image = cv2.imread("output_image.jpg")
png_image = cv2.imread("output_image.png")

jpeg_rgb = cv2.cvtColor(jpeg_image, cv2.COLOR_BGR2RGB)
png_rgb = cv2.cvtColor(png_image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(jpeg_rgb)
plt.title("JPEG")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(png_rgb)
plt.title("PNG")
plt.axis("off")

plt.tight_layout()
plt.show()
```

JPEG



PNG



```
gray_image = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2GRAY)

print("Grayscale image shape:", gray_image.shape)
```

```
Grayscale image shape: (1032, 1131)
```

```
hsv_image = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2HSV)

print("HSV image shape:", hsv_image.shape)
```

```
HSV image shape: (1032, 1131, 3)
```

```
lab_image = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2LAB)

print("LAB image shape:", lab_image.shape)
```

```
LAB image shape: (1032, 1131, 3)
```

```
new_width = width // 2
new_height = height // 2

resized_image = cv2.resize(
    image_bgr,
    (new_width, new_height)
)

resized_rgb = cv2.cvtColor(resized_image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(7, 5))
plt.imshow(resized_rgb)
plt.title("Resized Image - 50%")
plt.axis("off")
plt.show()

print("Original size:", image_bgr.shape)
print("Resized size :", resized_image.shape)
```




```
Original size: (1032, 1131, 3)
Resized size : (516, 565, 3)
```

```
rotated_image = cv2.rotate(
    image_bgr,
    cv2.ROTATE_90_CLOCKWISE
)

rotated_rgb = cv2.cvtColor(rotated_image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(7, 5))
plt.imshow(rotated_rgb)
plt.title("Rotated Image - 90° Clockwise")
plt.axis("off")
plt.show()
```




```
horizontal_flip = cv2.flip(image_bgr, 1)

horizontal_flip_rgb = cv2.cvtColor(
    horizontal_flip,
    cv2.COLOR_BGR2RGB
)

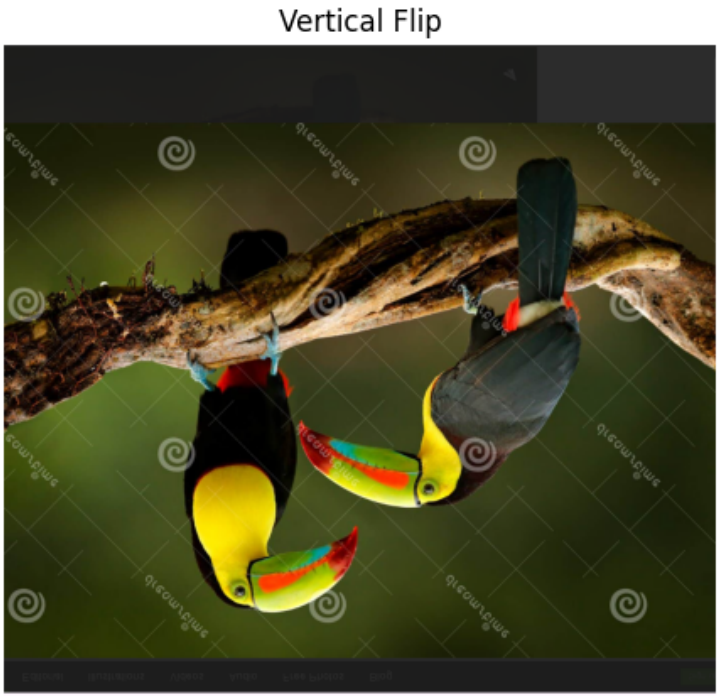
plt.figure(figsize=(7, 5))
plt.imshow(horizontal_flip_rgb)
plt.title("Horizontal Flip")
plt.axis("off")
plt.show()
```



```
vertical_flip = cv2.flip(image_bgr, 0)

vertical_flip_rgb = cv2.cvtColor(
    vertical_flip,
    cv2.COLOR_BGR2RGB
)

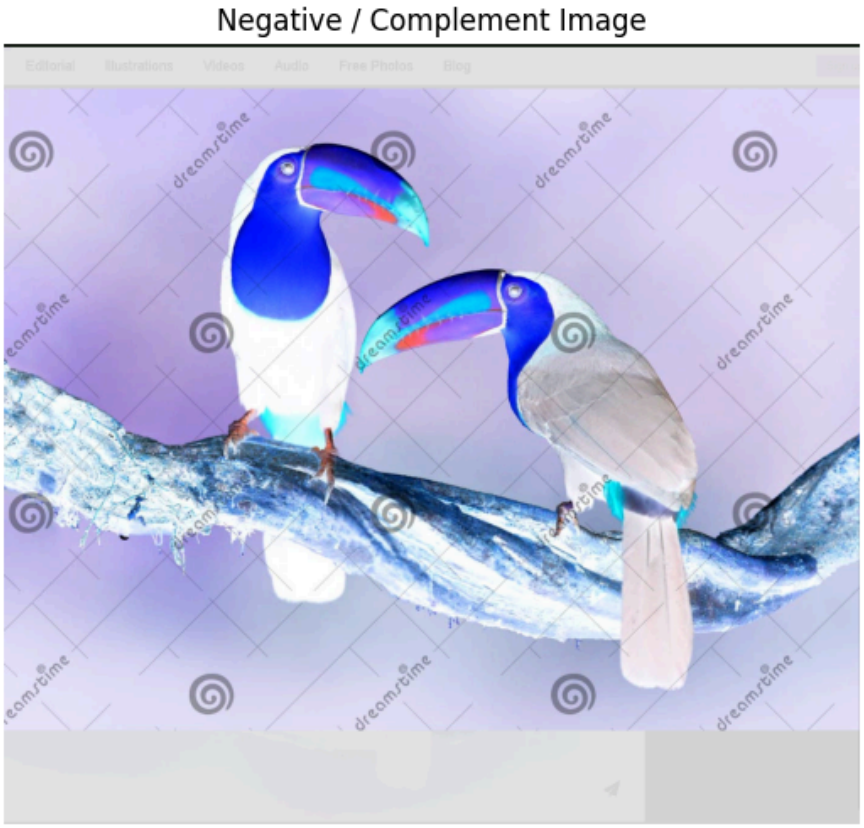
plt.figure(figsize=(7, 5))
plt.imshow(vertical_flip_rgb)
plt.title("Vertical Flip")
plt.axis("off")
plt.show()
```




```
negative_image = 255 - image_bgr

negative_rgb = cv2.cvtColor(
    negative_image,
    cv2.COLOR_BGR2RGB
)

plt.figure(figsize=(8, 6))
plt.imshow(negative_rgb)
plt.title("Negative / Complement Image")
plt.axis("off")
plt.show()
```




```
original_pixel = image_bgr[100, 100]
negative_pixel = negative_image[100, 100]

print("Original pixel :", original_pixel)
print("Negative pixel :", negative_pixel)
print("255 - Original :", 255 - original_pixel)
```

```
Original pixel : [ 4 50 38]
Negative pixel : [251 205 217]
255 - Original : [251 205 217]
```

```
# Define ROI dimensions
roi_width = width // 2
roi_height = height // 2

# Starting coordinates
x = (width - roi_width) // 2
y = (height - roi_height) // 2

# Crop ROI
roi = image_bgr[
    y:y + roi_height,
    x:x + roi_width
]

roi_rgb = cv2.cvtColor(roi, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(7, 5))
plt.imshow(roi_rgb)
plt.title("Region of Interest (ROI)")
plt.axis("off")
plt.show()
```

```
print("ROI Shape:", roi.shape)
```




```
ROI Shape: (516, 565, 3)
```

```
print("ROI Height:", roi.shape[0])
print("ROI Width :", roi.shape[1])
print("ROI Channels:", roi.shape[2])
print("ROI Data Type:", roi.dtype)
print("ROI Minimum Pixel:", roi.min())
print("ROI Maximum Pixel:", roi.max())
print("ROI Mean Pixel Value:", roi.mean())
```

```
ROI Height: 516
ROI Width : 565
ROI Channels: 3
ROI Data Type: uint8
ROI Minimum Pixel: 0
ROI Maximum Pixel: 255
ROI Mean Pixel Value: 64.14245043561776
```

```
plt.figure(figsize=(15, 12))

# Original
plt.subplot(3, 3, 1)
plt.imshow(image_rgb)
plt.title("Original")
plt.axis("off")

# Grayscale
plt.subplot(3, 3, 2)
plt.imshow(gray_image, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

# HSV
plt.subplot(3, 3, 3)
plt.imshow(cv2.cvtColor(hsv_image, cv2.COLOR_HSV2RGB))
```

```
plt.title("HSV")
plt.axis("off")

# LAB
plt.subplot(3, 3, 4)
plt.imshow(cv2.cvtColor(lab_image, cv2.COLOR_LAB2RGB))
plt.title("LAB")
plt.axis("off")

# Resized
plt.subplot(3, 3, 5)
plt.imshow(resized_rgb)
plt.title("Resized")
plt.axis("off")

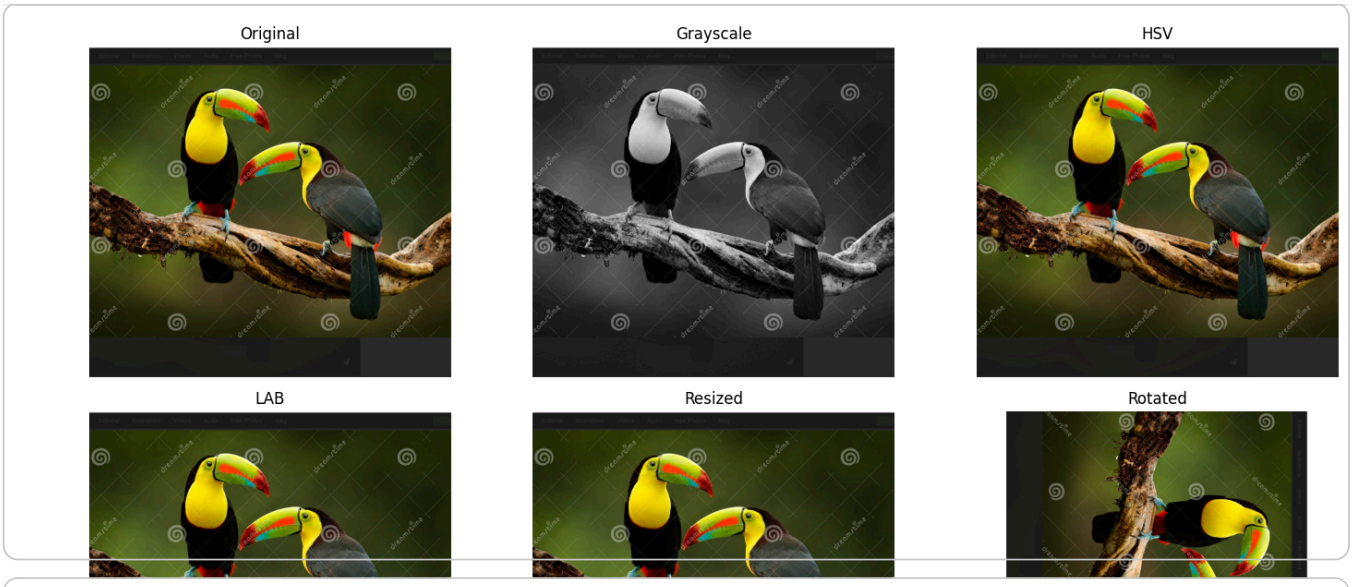
# Rotated
plt.subplot(3, 3, 6)
plt.imshow(rotated_rgb)
plt.title("Rotated")
plt.axis("off")

# Horizontal Flip
plt.subplot(3, 3, 7)
plt.imshow(horizontal_flip_rgb)
plt.title("Horizontal Flip")
plt.axis("off")

# Vertical Flip
plt.subplot(3, 3, 8)
plt.imshow(vertical_flip_rgb)
plt.title("Vertical Flip")
plt.axis("off")

# Negative
plt.subplot(3, 3, 9)
plt.imshow(negative_rgb)
plt.title("Negative")
plt.axis("off")

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
```